8.4 Visual Question Answering

- a VQA dataset containing $\sim$0.25M images, $\sim$0.76M questions, and $\sim$10M answers is constructed
- in order to perform VQA, a two-channel vision (image) + language (question) model is developed that culminates with a softmax over K possible outputs
- the top K = 1000 most frequent answers are chosen as possible outputs
- this set of answers covers 82.67% of the train+validation answers in the VQA dataset
- the different components of the model are described below
- the **Image Channel** provides an embedding for the image
- two types of embeddings are used:
  1. **I**: The activations from the last hidden layer of the VGG network are used as 4096-dimensional image embedding.
  2. **norm I**: These are ℓ_2-normalized activations from the last hidden layer of the VGG network.

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- the **Question Channel** provides an embedding for the question
- two types of embeddings are used:
  1. **LSTM Q**: An LSTM with one hidden layer is used to obtain a 1024-dimensional embedding for the question. The embedding obtained from the LSTM is a concatenation of last cell state and last hidden state representations (each being 512-dimensional) from the hidden layer of the LSTM. Each question word is encoded with a 300-dimensional embedding by a fully-connected layer + tanh nonlinearity which is then fed to the LSTM. The input vocabulary to the embedding layer consists of all the question words seen in the training dataset.
  2. **deeper LSTM Q**: An LSTM with two hidden layers is used to obtain a 2048-dimensional embedding for the question. The embedding obtained from the LSTM is a concatenation of the last cell state and last hidden state representations (each being 512-dimensional) from each of the two hidden layers of the LSTM. Hence the 2 (hidden layers) $\times$ 2 (cell state and hidden state) $\times$ 512 (dimensionality of each of the cell states, as well as hidden states) in Figure 10. This is followed by a fully-connected layer + tanh nonlinearity to transform the 2048-dimensional embedding to 1024 dimensions. The question words are encoded in the same way as in LSTM Q.

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- then, the image and question embeddings are combined to obtain a single embedding by the **Multi-Layer Perceptron (MLP)**
- for both the LSTM Q + I and deeper LSTM Q + norm I (Figure 10) methods, the image embedding is first transformed to 1024-dimensional by a fully-connected layer + tanh nonlinearity to match the LSTM embedding of the question
- the transformed image and LSTM embeddings (being in a common space) are then fused via element-wise multiplication
- this combined image + question embedding is then passed to an MLP – a fully-connected neural network classifier with 2 hidden layers and 1000 hidden units (dropout 0.5) in each layer with tanh nonlinearity, followed by a softmax layer to obtain a distribution over K answers
- the entire model is learned end-to-end with a cross-entropy loss
- the VGG network parameters are frozen to those learned for ImageNet classification and not fine-tuned in the image channel

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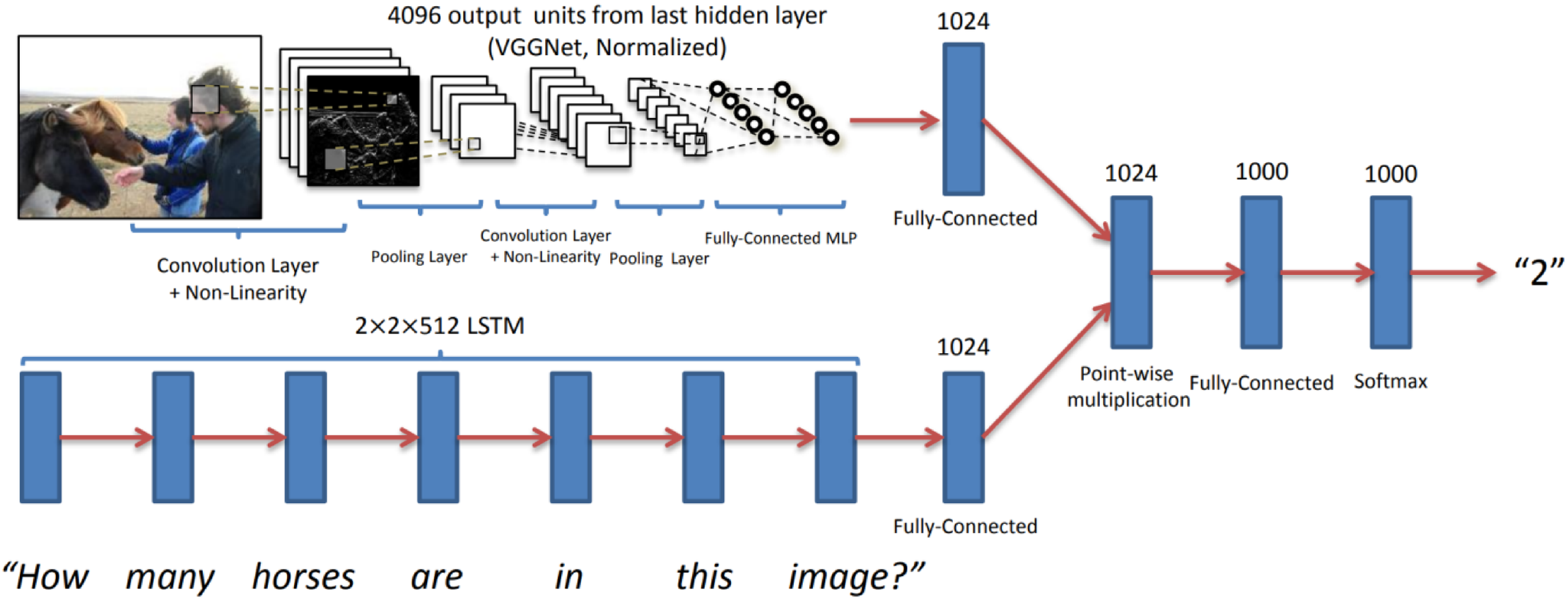


Figure 10: The best performing model (deeper LSTM Q + norm I). This model uses a two-layer LSTM to encode the questions and the last hidden layer of a VGG network to encode the images. The image features are then ℓ_2-normalized. Both the question and image features are transformed to a common space and fused via element-wise multiplication, which is then passed through a fully-connected layer followed by a softmax layer to obtain a distribution over answers.